DNA Extraction with kit

Quick Overview

Parameter	Information
Protocol ID	MOL-001
Version	1
Method Owner	Lieke Lipsch
Estimated Hands-on Time	4 hours
Total Duration	4 hours
Difficulty Level	☒ **Basic** ☐ Intermediate ☐ Advanced
Training Required	☒ **Yes** ☐ No
Biosafety Level	/
Typical Throughput	16
Sample types	Soil samples
Output	Extracted DNA, nanodrop values for DNA quality
Related Protocols	/

Method Information

Method Title: DNA extraction with kit

Author(s): Lieke Lipsch

Date Created: 22-01-2025

Last Updated: 22-01-2025

Reviewed By:

Approved By:

Purpose

What is this method used for? **Extracting DNA**

Scope

When should this protocol be used and when should it not be used?

Suitable for

- Soil samples

Not suitable for

- ?

Safety Considerations

Hazards

- Biological: /
- Chemical: /
- Physical: Make sure to use the right lid for the beads beater

Required PPE

- Lab coat
- Gloves

Waste Disposal

- Biological waste: tubes/filters/leftover samples - bag for autoclaving
- Chemical waste: washing buffer – add to non-halogenated waste bottles in fume hoods.
- Sharps: /

Materials and Reagents

Reagent/Material	Specification	Supplier	Storage
Nucleo Spin®	Soil	Machery- Nagel 740780.10 / .50 / .250 https://www.mn-net.com/de/nucleospin-soil-mini-kit-fuer-dna-aus-bodenproben-740780.50	Room temperature

Equipment

Equipment	Model	Location
Centrifuge with 30*2ml tube rotor Beads beater Nanodrop		DNA extraction lab 105 DNA extraction lab 105 DNA extraction lab 105

Procedure

Before Starting

Checklist

☐ Required lab training completed

☐ Required eplas training completed

☐ Kit ordered/ available

☐ Equipment and bench available/ reserved

☐ Sample IDs assigned

Step-by-Step Procedure

5 Protocol – purification of DNA from soil and sediment

Before starting the preparation: Clean bench

- Check Lysis Buffer SL1 or SL2 for precipitated SDS. Dissolve any precipitate by incubating the buffer at 30–40 °C for 10 min and shaking the bottle every 2 min.

Note down which samples you're analysing

1 Prepare sample

See section 2.4 and 2.5 for more information on the amount of starting material and the choice of lysis buffer. See section 2.7 for the repeated extraction of a sample to improve DNA yield.



250–500 mg sample +700 µL SL1 or SL2

400 mg

Transfer ~~**250–500 mg**~~ fresh **sample material** to a **MN Bead Tube Type A** containing the ceramic beads.

Important: **Do not fill** the tube **higher** than the **1 mL mark.**

Add **700 µL Buffer SL1** or ~~**Buffer SL2.**~~

Note for very dry material: If the sample material soaks up too much lysis buffer, fill the MN Bead Tube Type A up to the 1.5 mL mark with fresh lysis buffer.

Note for very wet material: Remove excess liquid before addition of lysis buffer, if necessary after spinning down the sample.

2 Adjust lysis conditions

Add **150 µL Enhancer SX** and close the cap.



+150 µL SX

Note: Enhancer SX ensures the highest possible DNA yield. It can, however, also promote the release of humic acids. See section 2.5 on how to lower the volume or omit the buffer entirely in order to increase DNA purity.

3 Sample lysis

~~*See section 2.6 for more information on homogenization methods (e.g., FastPrep® 24 instrument, Vortex adapter).*~~



~~**Vortex BT, 5 min**~~

~~Attach the MN Bead Tubes **horizontally** to a vortexer, for example, by taping or using a special adapter.~~

Fast prep/ beads beater 30sec, 5m/s

~~Vortex the samples at **full speed** and **room temperature** (18–25 °C) for **5 min**.~~

Get beaker for waste

4 Precipitate contaminants

Centrifuge for **2 min** at **11,000 x *g*** to eliminate the foam caused by the detergent.



11,000 x *g*, 2 min

Note: The clear supernatant can be transferred to a new collection tube (not provided) prior to the following precipitation. This might result in more consistent yields from prep to prep and is highly recommended for carbonate containing samples. See also section 2.7 for repeated extraction of a sample to improve DNA yield.

Add **150 µL Buffer SL3** and vortex for **5 s**. Hold 1 finger on top

+150 µL SL3

Vortex 5 s

Incubate for **5 min** at **0–4 °C**. In fridge, prepare/ label next step

0–4 °C, 5 min

Centrifuge for **1 min** at **11,000 x *g***.



11,000 x *g*, 1 min

5 Filter lysate

Place a **NucleoSpin® Inhibitor Removal Column** (red ring) in a Collection Tube (2 mL, lid).



Load supernatant

~650uL

Load up to **700 µL** clear supernatant of step 4 onto the filter.

Centrifuge for **1 min** at **11,000 x *g***.



11,000 x *g*, 1 min

Note: With very wet samples (e.g., sediments) the volume of clear supernatant of step 4 can exceed 700 µL significantly. In this case transfer the NucleoSpin® Inhibitor Removal Column to a new collection tube (not provided) and load the remaining supernatant. Centrifuge for 1 min at 11,000 x g. Combine the flow throughs.

Discard the NucleoSpin® Inhibitor Removal Column.

If a pellet is visible in the flow through, transfer the clear supernatant to a new collection tube (not provided).

6 Adjust binding conditions

Add **250 µL Buffer SB** and close the lid.



+250 µL SB

Vortex for **5 s**.

Vortex 5 s

Note: If samples were stabilized in Zymo DNA/RNA Shield, quantify total sample volume after addition of Buffer SB and add 0.2 volumes of isopropanol.

7 **Bind DNA**

Label soil column lid (green ring)

Place a **NucleoSpin® Soil Column** (green ring) in a Collection Tube (2 mL). Without lid!

Load **550 µL sample** onto the column.

Centrifuge for **1 min** at **11,000 x *g***.



Load 550 µL sample

11,000 x *g*, 1 min

Discard flow through and place the column back into the collection tube.

Load the **remaining sample** onto the column.

Centrifuge for **1 min** at **11,000 x *g***.



Load remaining sample

11,000 x *g*, 1 min

Discard flow through and place the column back into the collection tube.

8 Wash and dry silica membrane

1 st wash



+500 µL SB

Add **500 µL Buffer SB** to the NucleoSpin® Soil Column.

Centrifuge for **30 s** at **11,000 x *g***.



11,000 x *g*, 30 s

Discard flow through and place the column back into the collection tube.

2nd wash



+550 µL SW1

Add **550 µL Buffer SW1** to the NucleoSpin® Soil Column.

Centrifuge for **30 s** at **11,000 x *g***.

11,000 x *g*, 30 s

Discard flow through and place the column back into the collection tube.

3rd wash



+650 µL SW2

Vortex 2 s

Add **650 µL Buffer SW2** to the NucleoSpin® Soil Column.

Close the lid and vortex for **2 s**. Centrifuge for **30 s** at **11,000 x *g***. Discard flow through and place the column back into the collection tube.



11,000 x *g*, 30 s

4th wash



+650 µL SW2

Vortex 2 s

Add **650 µL Buffer SW2** to the NucleoSpin® Soil Column.

Close the lid and vortex for **2 s**. Centrifuge for **30 s** at **11,000 x *g***. Discard flow through and place the column back into the collection tube.



11,000 x *g*, 30 s

Note: The same collection tube is used throughout the entire washing procedure to reduce plastic waste. If new collection tubes are to be used for each step, see section 6.2 for ordering information.

9 Dry silica membrane



Centrifuge for **2 min** at **11,000 x *g***.

If for any reason, the liquid in the collection tube has touched the NucleoSpin® Soil Column after the drying step, discard flow through and centrifuge again.



11,000 x *g*, 5 min

10 Elute DNA

1.5ml microcentrifuge tubes, autoclaved

Place the NucleoSpin® Soil Column into a new microcentrifuge tube (not provided).



30–100 µL SE

Add ~~**30 µL** (for high concentration)~~, **50 µL** (for medium concentration and yield), ~~or **100 µL** (for high yield)~~ **Buffer SE** to the column.

Make sure this lands on the white filter paper, use new tip for each sample

RT, 1 min

Do not close the lid and incubate for 1 min at room temperature (18–25 °C). Close the lid and centrifuge for **30 s** at **11,000 x *g***.



11,000 x *g*, 30 s

Note: Quantify DNA not only by UV-VIS but also run an agarose gel to verify yield and DNA quality (see section 2.9 for more information).

Quantify DNA using the Nanodrop. Explanation see page 23.

Nanodrop:
1. turn on using button on back side
2. use program dsDNA
3. clean sensor with MQwater and a tissue
4. use SE buffer (from protocol above) as blank, close arm when ready to measure
5. after the blank clean sensor with tissue, and load 1,5uL sample
6. clean with tissue in between samples, and write down values
7. when all samples are done, click end experiment, and save data to a USB drive
8. clean sensor, and turn off using button
9. refill pipette tips

Critical Control Points

⚠ Step: /

Why it is critical:/

How to avoid mistakes:/

Quality Control

QC Parameter	Acceptance Criterion
Nanodrop	DNA can be measured

Troubleshooting

Problem	Possible Cause	Solution
No/low DNA concentration	Not ideal lysis conditions for your samples. Too low DNA concentration in your sample	Try the other lysis buffer (SL2) and try if you get better results without the enhancer SX. Try to concentrate your sample.

Expected Outcome

You should expect to have a measurable amount of DNA after the extraction.

Data Recording in Lab Books

Record the following information:

- Operator
- Date
- Sample IDs
- Batch numbers
- Deviations from protocol
- Raw data location

Storage and Stability

Item	Storage Condition	Stability
Extracted DNA	-20 degrees	Months

References

- SOPs: https://www.mn-net.com/media/4d/42/06/1793581373/Instruction-NucleoSpin-Soil_pa.pdf?ts=1793581373
- Publications
- Instrument manuals

Revision History

Version	Date	Changes	Author
1.0		Initial version	